



BSc (Hons) in Computer Science via GDSE

Module Code ITS1066

Professional Software Project in IT

Pallet Supply Management System

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1.0 Introduction to business

Pallets supply company is a small manufacturing business that produces wooden pallets to several companies to store some product and transport to other countries. The company mainly supplies these pallets to roofing sheets manufacturing company and coir manufacturing company.

At present, this company's daily operations, such as managing materials (like Wooden straps), nails stock, order handling, handling of supplies, dates of delivery of the pallets for related companies, and managing suppliers are done manually. As well as the owner will calculate the profit after the delivery of the stock in PO. but these fully manual system often cause many problems such as data duplication, Some orders are canceled due to late delivery times, some PO numbers are missed to the owner and immediately available quantities are unable to find and etc.

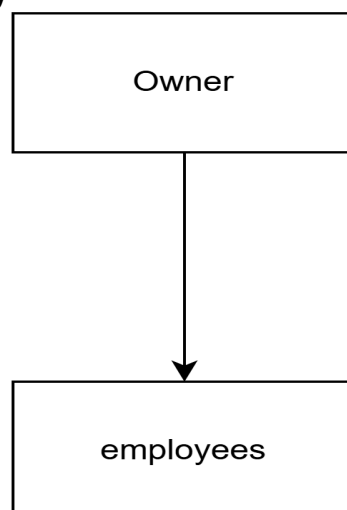
Business Model

The Pallet Supply Company operates on a manufacturing-based business model focused on the production and delivery of wooden pallets.

In addition to pallet production, The company manages employees, handling purchase orders, calculating profit after order completion, and maintaining stock of raw materials such as wooden straps, nails, and treatment chemicals.

The target market consists of manufacturing businesses that require strong, durable pallets for storage and transportation of goods. These include roofing sheet manufacturers, coir product factories, and other industrial suppliers who depend on consistent pallet availability for their operations. The company aims to provide reliable, timely, and cost-effective pallet supply services while maintaining long-term business relationships with its clients.

1.1 Employee hierarchy



2.0 Overview of the proposed solution

The proposed solution is an Automated Pallet Supply Management System designed to replace the existing manual workflow. The system will be developed using Java for the application and MySQL as the database management system.

.This solution addresses the following key needs:

Stock management

This system enables us to track and manage our wooden straps quantity,nails quantity,etc. it provides stock level in every day.allowing us to monitor product availability.then improve overall inventory control.

Employee Management

This solution includes features for employee management,such as how many pallets are created in one day. It helps to track employees' performance.

Customer Data Management

This system provide handle how many order quantity have for the owner,order date measure,owner can determine which customer send the order for the customer and etc.

Supplier Management

This system helps which supplier provide the items to the company as well as how many stock are bring the supplier at one time, measure payments of the suppliers and etc.

Profit management

This help to the owner when owner deliver the product to the customer after he can measure how many profit achieved in one order and etc.

Extra features

These provide for the additional features for the owner.this system provides a calculator and timber measurement calculation.

3.0 Advantages of the solution

1. Improved Efficiency
2. Reduced Human Errors
3. Real-Time Stock Visibility
4. Timely Delivery Management
5. Accurate Profit Calculation
6. Reduced Paperwork
7. Cost Savings
8. Professional Business Image
9. Enhanced Security

4.0 Use Case Diagram

